An *ab-initio* Diagrammatic Monte Carlo prototype for the LiF electron polaron

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Abstract

This note fixes the **goals**, deliberate **non-goals**, and **deferred** extensions of the `rmc-ab-initio` project: a diagrammatic Monte Carlo (DiagMC) prototype, built on the `ironmc` engine, that replaces the analytic Fröhlich electron–phonon vertex $|g(\mathbf{q})|^2 \propto \alpha/q^2$ with the *real* electron–phonon matrix elements $g(\mathbf{k}, \mathbf{q}, \nu)$ obtained from density-functional perturbation theory and Wannier interpolation (Quantum ESPRESSO + EPW). The first validation material is the electron polaron in LiF, a wide-gap ionic insulator that sits close to the textbook Fröhlich limit and therefore provides a built-in analytic cross-check. The scope below is the *first* prototype; Section 4 records what is intentionally postponed.

1 Overview

`ironmc` is a Rust DiagMC engine that samples the imaginary-time self-energy expansion of a polaron. Its only physics application today is the *analytic* Fröhlich model, whose vertex and dispersions are closed-form scalars. This project adds a second, *ab-initio* model: the same sampler, but with a material-specific electron–phonon (e–ph) vertex read from a real DFT/DFPT + Wannier calculation.

The data pipeline is

$$\underbrace{\text{pw.x, ph.x}}_{\text{Quantum ESPRESSO}} \longrightarrow \underbrace{\text{epw.x}}_{\text{Wannier e–ph tensors}} \longrightarrow \underbrace{\text{Rust reader + reconstruction}}_{\texttt{crates/apps/rmc-ab-initio}} \longrightarrow \text{DiagMC.}$$

DFPT supplies the phonons, the Born effective charges $\mathbf{Z}^*_\kappa$, and the electronic dielectric tensor $\boldsymbol{\varepsilon}_\infty$; EPW supplies the e–ph coupling in the localized Wannier representation $g(\mathbf{R}_e, \mathbf{R}_p)$, from which $g(\mathbf{k}, \mathbf{q}, \nu)$ is recovered by an inverse Fourier transform.

Why LiF. Rocksalt LiF has a single, non-degenerate conduction-band minimum at Γ, one dominant longitudinal-optical (LO) branch, and an intermediate-to-strong Fröhlich coupling $\alpha \approx 5$. It is therefore the cleanest first material: nearly the textbook Fröhlich case, with a genuine analytic limit to validate against.

Units. We work in Fröhlich units $\hbar = m^* = \omega_{\mathrm{LO}} = 1$. In these units the parabolic electron dispersion is exactly $p^2/2$ and the Einstein LO phonon is exactly 1, so the *only* material-specific quantity in the parabolic prototype is the vertex $|g(\mathbf{q})|^2$. The Fröhlich coupling is

$$\alpha \;=\; \frac{e^2}{\hbar}\sqrt{\frac{m^*}{2\hbar\omega_{\mathrm{LO}}}}\left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\right), \tag{1}$$

and energies are reported back in meV via $\hbar\omega_{\mathrm{LO}}$.

2 Goals

G1 — Real vertex. Drive the DiagMC sampler with the *ab-initio* e–ph vertex in place of the analytic one. The vertex is decomposed as

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; g^{\mathcal{S}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;+\; g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}), \tag{2}$$

the short-range (Wannier-interpolated) part plus the long-range polar part.

G2 — Engine untouched. Reuse the state-generic `ironmc` engine (`Runner`, `Update`, `MetropolisKernel`) *bit-for-bit*. In Fröhlich units the mass and phonon-energy constants never change, so the ab-initio model enters through a single vertex-weight seam rather than an engine rewrite.

G3 — Correct polar long range. Restore the long-range Fröhlich $1/q$ divergence that dominates the LiF polaron. EPW was run with `lpolar=.true.`, which *subtracts* the long-range part before Wannierization — from *both* the e-ph vertex and the dynamical matrix — so $g^{\mathcal{S}}$ from the tensor is short-ranged and both the vertex term $g^{\mathcal{L}}$ (Eq. (3)) and the dynamical-matrix term $D^{\mathcal{L}}$ (Eq. (4)) must be added back analytically. The latter is what restores the physical TO frequency and the LO–TO splitting; without it the phonon zero-point factor $(2M_\kappa\omega)^{-1/2}$ in $g^{\mathcal{L}}$ is mis-scaled (Section 6, P7). *Done:* both implemented as from-scratch ports of EPW's `rgd_blk_epw` and `rgd_blk` (Ewald parameters $\Lambda_{\mathrm{Ew}} = 1$, $g_{\max} = 14$, $e^2 = 2$ Ry) and validated to machine precision against the coarse Bloch `.epb` and dynamical matrix `dynq` (Section 6, P6–P7).

G4 — Clean Rust reader. Implement the EPW-file readers and the Wannier reconstruction from scratch in Rust, with no Python at runtime. The existing Python post-processor (`patt126-ab-initio-diagmc`) is used as a *reference only*, not copied; the rewrite also fixes a latent index assumption in the reference.

G5 — Two-front validation. Benchmark the LiF electron polaron against *both* (i) Fröhlich-limit DiagMC energetics at α_{LiF} (Mishchenko *et al.*, 2000), and (ii) *ab-initio* LiF polaron numbers (Sio, Verdi, Poncé, Giustino, 2019).

The long-range term used for G3 (Verdi–Giustino form) is

$$g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; i\,\frac{4\pi e^2}{\Omega}\sum_{\kappa}\left(\frac{\hbar}{2M_\kappa\omega_{\nu\mathbf{q}}}\right)^{1/2}\sum_{\mathbf{G}\neq-\mathbf{q}}\frac{(\mathbf{q}+\mathbf{G})\cdot\mathbf{Z}^*_\kappa\cdot\mathbf{e}_{\kappa\nu}(\mathbf{q})}{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}\,e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_\kappa}\,e^{-\frac{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}{4\Lambda}}\,\big[U_{\mathbf{k}+\mathbf{q}}U^\dagger_{\mathbf{k}}\big]_{mn}, \tag{3}$$

with Ω the cell volume, $M_\kappa, \boldsymbol{\tau}_\kappa$ the atomic masses and positions, $\mathbf{e}_{\kappa\nu}$ the phonon eigenvectors, and Λ an Ewald range-separation parameter. The final factor $[U_{\mathbf{k}+\mathbf{q}}U^\dagger_{\mathbf{k}}]_{mn}$ is the electron-band rotation into the smooth (Wannier) gauge; at the Γ-slice ($\mathbf{k} = \Gamma$, single conduction band) it reduces to the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}}|u_\Gamma\rangle$. Restoring it is essential to the exact short/long decomposition of Eq. (2); see Section 6, P6. As $q \to 0$ the term behaves as $1/q$, reproducing the Fröhlich tail $|g_F(\mathbf{q})|^2 \propto \alpha/q^2$.

The *same* dipole physics enters the dynamical matrix: EPW likewise removes the rigid-ion (dipole–dipole) long-range from D before Wannierization, so the interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical frequency with no LO–TO splitting. The term to add back (Gonze form, EPW `rgd_blk`) is, with $\mathbf{Q} \equiv \mathbf{q} + \mathbf{G}$,

$$D^{\mathcal{L}}_{\kappa\alpha,\kappa'\beta}(\mathbf{q}) \;=\; \frac{4\pi e^2}{\Omega}\sum_{\mathbf{G}}\frac{e^{-\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}/4\Lambda}}{\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}}\,(\mathbf{Q}\cdot\mathbf{Z}^*_\kappa)_\alpha\,(\mathbf{Q}\cdot\mathbf{Z}^*_{\kappa'})_\beta\,e^{i\mathbf{Q}\cdot(\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})} \;-\; \delta_{\kappa\kappa'}\,S_{\kappa\alpha\beta}, \tag{4}$$

where the self-term $S_{\kappa\alpha\beta} = \sum_{\kappa''} D^{\mathcal{L}}_{\kappa\alpha,\kappa''\beta}(\mathbf{0})$ (the same sum at $\mathbf{q} = 0$, summed over the second atom) is subtracted on the diagonal blocks to enforce the acoustic sum rule. Diagonalizing $D^{\mathcal{S}} + D^{\mathcal{L}}$ (mass-reduced) gives the physical branches $\omega_{\mathbf{q}\nu}$ and eigenvectors $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ that enter Eq. (3).

3 Non-goals (deliberate scope of the first prototype)

These are conscious simplifications, chosen so the prototype is small, exact where it matters, and cheaply validated. Each is lifted in Section 4.

N1 — Electron polaron only. The single, non-degenerate conduction-band minimum at Γ. The hole/valence polaron (degenerate top valence band) is out of scope.

N2 — Γ-slice vertex. We approximate $g(\mathbf{k}, \mathbf{q}) \approx g(\Gamma, \mathbf{q})$, i.e. the vertex depends on the phonon momentum $\mathbf{q}$ only, evaluated as if the electron sits at the band bottom. True electron-momentum dependence is a non-goal here.

N3 — One Einstein LO branch. Keep only the Fröhlich-active LO branch and treat it as dispersionless at ω_{LO}. The remaining branches and the dispersion $\omega_{\mathrm{LO}}(\mathbf{q})$ are not modeled.

N4 — No tensor compression. The reduced object the sampler needs is the scalar field $|g(\Gamma, \mathbf{q}, \mathrm{LO})|^2$, which is small on any grid. It is precomputed once and interpolated; *no* SVD/Tucker low-rank compression is used. (Fixing $\mathbf{k} = \Gamma$ is itself an exact, lossless dimensional reduction that supersedes lossy compression in this regime.)

N5 — Reweighting, not native sampling. The ab-initio vertex is introduced by reweighting the existing analytic Fröhlich importance-sampling envelope, not by a native $|g_{\mathrm{ab}}(\mathbf{q})|^2$ momentum sampler.

N6 — No engine parameterization. The engine's mass/phonon-energy constants stay fixed at unity (Fröhlich units); no `MASS`/`OMEGA` genericization is introduced for the parabolic prototype.

N7 — No GPU path. The ab-initio model targets the CPU driver only; the GPU (`#[cube]`) mirror is not extended to it.

N8 — Single material. LiF is the sole validation target; this is not yet a general multi-material framework.

4 Deferred to future work

Listed roughly in the order they would be relaxed. None of these require reworking the Phase-A readers or reconstruction; they extend the same pipeline.

D1 — Full $g(\mathbf{k}, \mathbf{q})$ k-dependence. Relaxing the Γ-slice (N2) turns the vertex into a function on the six-dimensional $(\mathbf{k}, \mathbf{q})$ space, which cannot be precomputed onto a grid. This is exactly the regime where **SVD/Tucker compression + on-the-fly evaluation** of the Wannier tensor becomes necessary (reversing N4). Relevant at strong coupling, where the electron explores momenta away from Γ.

D2 — Multi-branch, dispersive phonons. Include all phonon branches and the momentum dependence $\omega_\nu(\mathbf{q})$, replacing the single Einstein LO branch (N3).

D3 — Non-parabolic electrons. A non-parabolic $\varepsilon(\mathbf{k})$ requires genericizing the engine's dispersion (the deferred `MASS`/`OMEGA` parameterization and `Model`-trait seam), lifting N6.

D4 — Native q-sampler. A momentum sampler drawn directly from $|g_{\mathrm{ab}}(\mathbf{q})|^2$, replacing the reweighting of N5 for sampling efficiency.

D5 — Hole polaron. The degenerate valence-band-maximum polaron (lifting N1), which needs a multi-band, degenerate treatment.

D6 — GPU path. A `#[cube]` GPU implementation of the ab-initio model (lifting N7), guarded by the existing lockstep parity tests.

D7 — More materials. Generalize beyond LiF once the LiF prototype is validated (lifting N8).

5 Validation ladder

The goals of Section 2 are checked in three rungs of increasing physical content:

1. **Refactor parity.** The Fröhlich model, re-expressed through the new ab-initio code path, reproduces the existing engine's $\alpha = 1$ benchmark ($E \approx -1.016$, $Z \approx 0.59$).
2. **Fröhlich-limit + unit bridge.** Fed the *analytic* $|g(\mathbf{q})|^2 \propto \alpha_{\mathrm{LiF}}/q^2$, the ab-initio model must match the pure Fröhlich engine at α_{LiF} against Mishchenko *et al.* (2000); and the *real* tensor's $|g(\Gamma, \mathbf{q}\to 0, \mathrm{LO})|^2$ must recover the same $1/q^2$ asymptote.
3. **Material comparison.** The real LiF vertex yields a polaron formation energy and effective mass compared to Sio *et al.* (2019).

6 Practical pitfalls and fixes (A0/A1)

Turning the EPW run into a *usable* vertex meant working through several failures whose symptoms were misleading. They are recorded here so they are not rediscovered; the blow-by-blow diagnosis, with reproducing commands, lives in `datasets/LiF/EPH-NAN-DIAGNOSIS.md`.

P1 — Ultrasoft pseudopotentials give an all-`NaN` e–ph tensor (root cause). The first A0 deck used ultrasoft pseudopotentials (SSSP-efficiency GBRV). EPW's electron–phonon matrix elements are only supported with *norm-conserving* pseudopotentials — the ultrasoft augmentation charge is not carried into the coupling. The coupling came out finite *only* at $\mathbf{q} = \Gamma$ (the trivial reference q, which needs no symmetry rotation) and `NaN` at every other $\mathbf{q}$; the Bloch→Wannier sum then spread the `NaN` to 100% of `LiF.epmatwp`. Because `ph.x` *does* support ultrasoft, the phonons, dynamical matrix, $\mathbf{Z}^*_\kappa$ and $\boldsymbol{\varepsilon}_\infty$ were all correct — only the e–ph coupling was wrong, which is why the fault was easy to misattribute. *Fix:* switch Li and F to norm-conserving Pseudo Dojo ONCV (scalar-relativistic PBE). This raises the required plane-wave cutoff (`ecutwfc` : 50 → 96 Ry, `ecutrho` : 400 → 384 Ry, i.e. the norm-conserving 4×), which makes `ph.x` ∼3–4× slower (the wavefunctions move from the small smooth FFT grid onto the dense grid) — an unavoidable cost of the correct pseudopotential family. With norm-conserving pseudos the tensor is finite at every $\mathbf{q}$, and $\boldsymbol{\varepsilon}_\infty$, $\mathbf{Z}^*_\kappa$ are essentially unchanged.

P2 — "Right size, wrong values": a silent failure. The broken run finished cleanly and wrote `LiF.epmatwp` at exactly the expected byte size; the only hint was an easily-dismissed `IEEE_INVALID` note at exit. The file was 100% uninitialised-memory `NaN` (a uniform garbage word that even *changed* between reruns). An earlier "A0 done" had been declared on file size alone, without ever checking the tensor *values*. *Fix:* the pipeline (`run_pipeline.nu`) now runs an explicit finiteness check on `LiF.epmatwp` and aborts loudly on `NaN`, so a broken run can no longer masquerade as a good one.

P3 — Two dead ends from assuming an upstream bug. Before the pseudopotential cause was found, two "EPW bug" hypotheses were pursued and *both failed*: (i) widening the Wannier outer window `dis_win_max` to enclose every band made the coarse tensor *worse*, not better; (ii) a bounds-guard patch to EPW's `gmap_sym` routine (a genuine latent out-of-bounds read) compiled in cleanly yet left the output byte-identical — so that read was not the cause. The lesson, recorded explicitly: suspect the *inputs* (here, an unsupported pseudopotential family) before patching upstream code.

P4 — The reference `.epb` reader has the wrong record layout. The Python reference (`patt126`) reads the coarse Bloch `.epb` file assuming a single band count throughout the record. The real record interleaves a *20-band* eigenvalue array with a *15-band* coupling array (of which only a 5×5 block is populated); a naïve parse silently mis-aligns and yields a *false positive* — a "the coarse data looks fine" reading that briefly sent the debugging down a wrong path. *Fix:* the true record layout was reverse-engineered and byte-verified — a *single* unformatted Fortran record (4-byte length markers) $[\, n_{qc},\ \mathbf{x}_{qc}(3, n_{qc}),\ \varepsilon(20, n_k),\ D(6, 6, n_{qc}),\ g(15, 15, n_k, 6, n_{qc}),\ \mathbf{Z}^*, \boldsymbol{\varepsilon}_\infty \,]$, all little-endian and column-major, one file per pool with the pool-local n_k recovered from the record length. Two indexing traps: the coupling's mode axis sits *interior* (stride $n_k \cdot 6$, not $6 \cdot n_k$), and the conduction-band minimum is kept-band local index 0 in the coupling array but eigenvalue index 5 in ε. The Rust reader uses this directly and decodes only the needed conduction-band slice on demand.

P5 — epw coarse-grid prerequisites (reproducibility). Several setup constraints are load-bearing and non-obvious: the `nscf` k-points must be an *explicit* Γ-centred crystal list (an automatic, half-shifted Monkhorst–Pack mesh breaks EPW's $\mathbf{k}{+}\mathbf{q}$ folding); the whole `scf/ph/nscf/epw` chain must use *one* Quantum ESPRESSO version (a mixed-version `phsave` corrupts the heap); and EPW needs a large stack (`ulimit -s unlimited`). Finally, changing the pseudopotentials shifts the band edges ($\sim$0.15 eV here), so the disentanglement windows (`dis_froz_max`, `dis_win_max`) must be re-tuned against the new `nscf` eigenvalues — otherwise the frozen conduction-band minimum, which must stay exact, can slip out of its window.

P6 — The polar term lives in the Cartesian displacement basis, and carries an electron-band rotat
Two subtleties in Eq. (3) decide whether the split $g = g^{\mathcal{S}} + g^{\mathcal{L}}$ of Eq. (2) actually closes, and both were found by round-tripping against the coarse `.epb` (which stores the *full*, pre-subtraction coupling). *(i) Basis.* EPW subtracts $g^{\mathcal{L}}$ from the coupling in the *Cartesian atomic-displacement* basis ($\kappa\alpha$, i.e. per atom and Cartesian direction), *before* the phonon-eigenvector projection $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ and the $(\hbar/2M_\kappa\omega)^{1/2}$ mode factor. So the reconstructed short-range part and the re-added $g^{\mathcal{L}}$ must both be formed in that basis and summed there; projecting onto branches first is inconsistent. The mode factor and eigenvector projection are applied afterwards by the consumer. *(ii) Gauge.* The band-rotation factor $[U_{\mathbf{k}+\mathbf{q}} U_{\mathbf{k}}^\dagger]_{mn}$ — at the Γ-slice the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}} | u_\Gamma \rangle$ — is easy to drop under the "vertex depends on $\mathbf{q}$ only" reduction (both electron legs rotated with U_Γ, overlap $\equiv 1$). Dropping it puts

the polar term in the wrong electron gauge: near Γ the reduced sum still overshoots $\|g\|$ by $\sim 9\%$, and the overlap itself falls from ≈ 0.94 at the first coarse shell to 0 at the zone boundary as the conduction-band character changes with $\mathbf{k}$. Restoring the exact bra ($u_{\Gamma+\mathbf{q}}$) and the overlap makes the decomposition exact. *Result:* against all 215 non-zero coarse $\mathbf{q}$, $g_{\text{full}} = g^{\mathcal{S}} + \langle u_{\Gamma+\mathbf{q}} | u_{\Gamma} \rangle \, g^{\mathcal{L}}$ to machine precision — gauge-invariant cosine = 1.000000 and $|\text{ratio} - 1| \sim 3 \times 10^{-7}$ (the `.epb` Kohn–Sham states carry an arbitrary per-$\mathbf{k}$ phase, so the check is on direction and magnitude, not the raw complex difference). The reduced $\mathbf{q}$-only vertex remains the object the sampler consumes; the exact bra + overlap is used only to prove the reconstruction and the polar term against real data.

P7 — The dynamical matrix carries the same polar long-range. The subtraction of P6 is not confined to the e-ph vertex: EPW removes the rigid-ion dipole–dipole term from the dynamical matrix as well (`rgd_blk`), so the Wannier-interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical 11.95 THz with no LO–TO splitting, where the physical TO is 8.12 THz (agreeing with `ph.x`). This is easy to overlook because the phonon *eigenvectors* are barely affected (the optical modes are symmetry-fixed Li–F relative motion), but the *frequency* enters the vertex through the zero-point factor $(2M_\kappa \omega_{\mathbf{q}\nu})^{-1/2}$, so a short-range ω mis-scales $|g|^2$ by a factor of order two. *Fix:* port EPW's `rgd_blk` (Gonze form, Eq. (4)) – the $\mathbf{q} \neq 0$ dipole–dipole sum over all atom-pair blocks plus the $\mathbf{q}$=0 self-term on the diagonal blocks – and add it back before diagonalizing. *Validated* at two levels: the reconstructed long-range term equals coarse-`dynq` minus the short-range force constants *element-by-element* (ratio 1.000), and all six branches then reproduce the DFPT `dynq` at the coarse q ($[2.76, 2.76, 5.23, 8.12, 8.12, 18.45]$ THz, LO = 18.45) — the phonon analog of the P6 round-trip. The restored $\omega_{\text{LO}}(\Gamma) = 18.67$ THz gives a Lyddane–Sachs–Teller ratio $\varepsilon_0/\varepsilon_\infty = (\omega_{\text{LO}}/\omega_{\text{TO}})^2 \approx 5.3$, i.e. $\varepsilon_0 \approx 11$ (experiment ≈ 9).